Vulnerability Scanning

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1. Vulnerability Scanning Lab

1.1 Introduction

This report presents the initial results of vulnerability scanning conducted on a test environment using tools such as Nmap, Nuclei, and Nikto. The objective is to identify security weaknesses and assess their severity using CVSS scores.

1.2 Methodology

- Service/version scan executed using Nmap:
  nmap -sV 192.168.56.101
- Vulnerability scanning performed using Nuclei with templates.
- Web server analysis conducted with Nikto.
- Results recorded and categorized based on CVSS scoring.

1.3 Tools used:

- Nmap
- Nuclei
- Nikto

1.4 Target ip: 192.168.56.101 (Metasploitable)

- **Nmap scan Output:**
  Command line ~ (*nmap -sV -A 192.168.56.101 -oN nmap.txt***)**
- **Nuclei Command and Output:**
  Command line ~ (*nuclei -u http://192.168.56.101 -severity medium,high,critical -o nuclei.txt***)**
- **Nikto scan output:**
  Command ~ (*perl nikto.pl -h http://192.168.56.101 -o nikto.txt***)**

1.5 Findings:

- Critical Vulnerabilities
  - CVE-2011-2523
  - CVE-2020-1938
  - CVE-2012-1823
  - PostgreSQL exposed with empty/default credentials

  These allow:
  - Remote code execution
  - Unauthorized system access
  - Full database compromise
- High Vulnerabilities
  - Outdated Apache (2.2.8)
  - Outdated PHP (5.2.4)
  - VNC weak credentials
  - Exposed phpMyAdmin
- Medium Vulnerabilities
  - Directory listing enabled

- Missing security headers
- HTTP TRACE enabled (XST attack risk)
- phpinfo() exposed

1.6 Remediation:

- Immediate Fixes
  - Patch all CVEs (Apache, PHP, FTP, Tomcat)
  - Remove default credentials (PostgreSQL)
  - Disable unused services (FTP, VNC)
- System Hardening
  - Upgrade Apache & PHP
  - Restrict phpMyAdmin access
  - Disable directory indexing
- Security Improvements
  - Disable TRACE method
  - Add security headers:
  - Content-Security-Policy
  - Strict-Transport-Security
  - X-Content-Type-Options